

Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y telecomunicaciones - sección 1

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REDES Y TELECOMUNICACIONES:

LABORATORIO 1

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Guatemala, 17 de febrero de 202

Topología de red utilizada

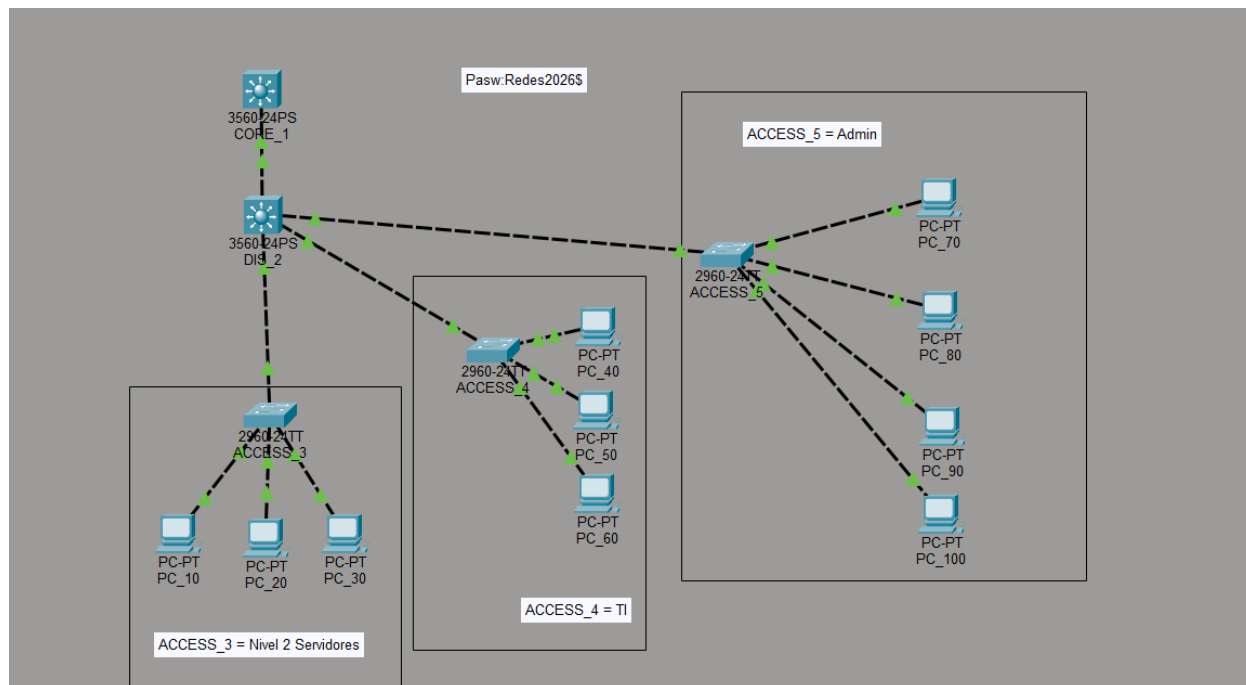


Tabla de IPS utilizadas

VLAN	Nombre	Área / Departamento	Red Asignada	Gateway (SVI)
10	Gerencia	Administración	10.10.10.0/24	10.10.10.1
20	Ventas	Comercial	10.20.20.0/24	10.20.20.1
30	RRHH	Recursos Humanos	10.30.30.0/24	10.30.30.1
40	Finanzas	Contabilidad	10.40.40.0/24	10.40.40.1
50	IT	Tecnología	10.50.50.0/24	10.50.50.1
60	Soporte	Técnico	10.60.60.0/24	10.60.60.1
70	Marketing	Publicidad	10.70.70.0/24	10.70.70.1
80	Logística	Operaciones	10.80.80.0/24	10.80.80.1
90	Seguridad	CCTV / Seguridad	10.90.90.0/24	10.90.90.1
100	Servidores	DataCenter	10.100.100.0/24	10.100.100.1

Configuración de dominio y creación de Vlans

Todo esto paso en CORE_1

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname CORE_1
CORE_1(config)#vtp mode server
Device mode already VTP SERVER.
CORE_1(config)#domain EMPRESA_X
^
% Invalid input detected at '^' marker.

CORE_1(config)#vtp domain EMPRESA_X
Changing VTP domain name from NULL to EMPRESA_X
CORE_1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to EMPRESA_X.

CORE_1(config)#vtp password cisco
Setting device VLAN database password to cisco
CORE_1(config)#version 2
^
% Invalid input detected at '^' marker.

CORE_1(config)#vtp version 2
CORE_1(config)#vlan 10
CORE_1(config-vlan)#name PRODUCCION
CORE_1(config-vlan)#vlan 20
CORE_1(config-vlan)#name DESARROLLO
CORE_1(config-vlan)#vlan 30 QA
^
% Invalid input detected at '^' marker.

CORE_1(config-vlan)#vlan 30
CORE_1(config-vlan)#name QA
CORE_1(config-vlan)#vlan 40
CORE_1(config-vlan)#name INFRA
CORE_1(config-vlan)#vlan 50
CORE_1(config-vlan)#name
CORE_1(config-vlan)#name APP
CORE_1(config-vlan)#vlan 60
CORE_1(config-vlan)#CIBER
^
% Invalid input detected at '^' marker.

CORE_1(config-vlan)#vlan 60
CORE_1(config-vlan)#name CIBER
CORE_1(config-vlan)#vlan 70
CORE_1(config-vlan)#name FINANZAS
CORE_1(config-vlan)#vlan 80
CORE_1(config-vlan)#name AUDITORIA
CORE_1(config-vlan)#vlan 90 MARKETING
^
% Invalid input detected at '^' marker.

CORE_1(config-vlan)#vlan 90
CORE_1(config-vlan)#MARKETING
```

Todas las vlans están activas y esta echo de manera truncal para evitar problemas por cuestiones de uso de Switch a Switch en el CORE y DIS se usan los puertos Gigabit

```

Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40,50,60,70,80,90,100
Gig0/2    1,10,20,30,40,50,60,70,80,90,100

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40,50,60,70,80,90,100
Gig0/2    10,20,30,40,50,60,70,80,90,100

CORE_1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES unset   down        down
FastEthernet0/2 unassigned      YES unset   down        down
FastEthernet0/3 unassigned      YES unset   down        down
FastEthernet0/4 unassigned      YES unset   down        down
FastEthernet0/5 unassigned      YES unset   down        down
FastEthernet0/6 unassigned      YES unset   down        down
FastEthernet0/7 unassigned      YES unset   down        down
FastEthernet0/8 unassigned      YES unset   down        down
FastEthernet0/9 unassigned      YES unset   down        down
FastEthernet0/10 unassigned      YES unset   down        down
FastEthernet0/11 unassigned      YES unset   down        down
FastEthernet0/12 unassigned      YES unset   down        down
FastEthernet0/13 unassigned      YES unset   down        down
FastEthernet0/14 unassigned      YES unset   down        down
FastEthernet0/15 unassigned      YES unset   down        down
FastEthernet0/16 unassigned      YES unset   down        down
FastEthernet0/17 unassigned      YES unset   down        down
FastEthernet0/18 unassigned      YES unset   down        down
FastEthernet0/19 unassigned      YES unset   down        down
FastEthernet0/20 unassigned      YES unset   down        down
FastEthernet0/21 unassigned      YES unset   down        down
FastEthernet0/22 unassigned      YES unset   down        down
FastEthernet0/23 unassigned      YES unset   down        down
FastEthernet0/24 unassigned      YES unset   down        down
GigabitEthernet0/1 unassigned      YES unset   up          up
GigabitEthernet0/2 unassigned      YES unset   up          up
Vlan1          unassigned      YES unset   administratively down down
Vlan10         10.10.10.1      YES manual  up          up
Vlan20         10.10.20.1      YES manual  up          up
Vlan30         10.10.30.1      YES manual  up          up
Vlan40         10.10.40.1      YES manual  up          up
Vlan50         10.10.50.1      YES manual  up          up
Vlan60         10.10.60.1      YES manual  up          up
Vlan70         10.10.70.1      YES manual  up          up
Vlan80         10.10.80.1      YES manual  up          up
Vlan90         10.10.90.1      YES manual  up          up
Vlan100        10.10.100.1     YES manual  up          up
CORE_1#

```

Se copian los accesos de el core al DIS para que las vilans estén con la misma configuración

```
DIS_2#  
DIS_2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	PRODUCCION	active	
20	DESARROLLO	active	
30	QA	active	
40	INFRA	active	
50	APP	active	
60	CIBER	active	
70	FINANZAS	active	
80	AUDITORIA	active	
90	MARKETING	active	
100	RRHH	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	

```
DIS_2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	PRODUCCION	active	
20	DESARROLLO	active	
30	QA	active	
40	INFRA	active	
50	APP	active	
60	CIBER	active	
70	FINANZAS	active	
80	AUDITORIA	active	
90	MARKETING	active	
100	RRHH	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	

--More-- |

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```

DIS_2#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : EMPRESA_X
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0007.EC2A.ED00
Configuration last modified by 0.0.0.0 at 3-1-93 09:24:02

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 1005
Number of existing VLANs : 15
Configuration Revision   : 41
MD5 digest               : 0x79 0x49 0x16 0xC1 0x2D 0x52 0x59 0x08
                        : 0xF5 0x26 0x1A 0x52 0xCB 0xCE 0xEE 0x15
DIS_2#

```

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Routing y ping entre maquinas mediante switchches

```

Pinging 10.10.30.10 with 32 bytes of data:
Reply from 10.10.30.10: bytes=32 time=4ms TTL=128
Reply from 10.10.30.10: bytes=32 time=4ms TTL=128
Reply from 10.10.30.10: bytes=32 time=5ms TTL=128
Reply from 10.10.30.10: bytes=32 time=4ms TTL=128

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 5ms, Average = 4ms

C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time=9ms TTL=127
Reply from 10.10.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
C:\>

```

Pingueando maquinas de Switch 4

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Ping statistics for 10.10.40.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.60.1

Pinging 10.10.60.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.60.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.40.1

Pinging 10.10.40.1 with 32 bytes of data:

Reply from 10.10.40.1: bytes=32 time<1ms TTL=255
Reply from 10.10.40.1: bytes=32 time<1ms TTL=255
Reply from 10.10.40.1: bytes=32 time=1ms TTL=255
Reply from 10.10.40.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.10.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.50.1

Pinging 10.10.50.1 with 32 bytes of data:

Reply from 10.10.50.1: bytes=32 time<1ms TTL=255
Reply from 10.10.50.1: bytes=32 time<1ms TTL=255
Reply from 10.10.50.1: bytes=32 time=10ms TTL=255
Reply from 10.10.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Conexiones de switch 5

```
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.70.10

Pinging 10.10.70.10 with 32 bytes of data:

Reply from 10.10.70.10: bytes=32 time=4ms TTL=128
Reply from 10.10.70.10: bytes=32 time=2ms TTL=128
Reply from 10.10.70.10: bytes=32 time=5ms TTL=128
Reply from 10.10.70.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms

C:\>ping 10.10.80.10

Pinging 10.10.80.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.80.10: bytes=32 time<1ms TTL=127
Reply from 10.10.80.10: bytes=32 time=11ms TTL=127
Reply from 10.10.80.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.80.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>ping 10.10.90.10

Pinging 10.10.90.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.90.10: bytes=32 time=1ms TTL=127
Reply from 10.10.90.10: bytes=32 time<1ms TTL=127
Reply from 10.10.90.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.90.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.100.10

Pinging 10.10.100.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.100.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Mescla de ambos puertos gigabit para la port-channel

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

CORE_1(config-if)#end
CORE_1#
%SYS-5-CONFIG_I: Configured from console by console

CORE_1#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          LACP       Gig0/1(P) Gig0/2(P)
CORE_1#show interface trunk
Port      Mode          Encapsulation  Status      Native vlan
Po1       on            802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       1-1005

Port      Vlans allowed and active in management domain
Po1       1,10,20,30,40,50,60,70,80,90,100

Port      Vlans in spanning tree forwarding state and not pruned
Po1       1,10,20,30,40,50,60,70,80,90,100

CORE_1#

```

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Evidencia de seguridad en switches

```
CORE_1(config)#ip ssh version 2
*Mar 1 14:37:54.283: %SSH-5-ENABLED: SSH 1.99 has been enabled
CORE_1(config)#
CORE_1(config)#line vty 0 15 transport input ssh
                                     ^
% Invalid input detected at '^' marker.

CORE_1(config)#line vty 0 15
CORE_1(config-line)#transport input ssh
CORE_1(config-line)#login local
CORE_1(config-line)#exit
CORE_1(config)#end
CORE_1#
%SYS-5-CONFIG_I: Configured from console by console

CORE_1#wr
Building configuration...
[OK]
CORE_1#write memory
Building configuration...
[OK]
CORE_1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
CORE_1#
```

```
DIS_2(config)#ip ssh version 2
*Mar 1 14:42:2.752: %SSH-5-ENABLED: SSH 1.99 has been enabled
DIS_2(config)#
DIS_2(config)#line vty 0 15
DIS_2(config-line)#transport input ssh
DIS_2(config-line)#login local
DIS_2(config-line)#exit
DIS_2(config)#exit
DIS_2#
%SYS-5-CONFIG_I: Configured from console by console

DIS_2#wr
Building configuration...
[OK]
DIS_2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
DIS_2#
```

```
ACCESS_3(config)#ip ssh version 2
*Mar 1 14:45:2.950: %SSH-5-ENABLED: SSH 1.99 has been enabled
ACCESS_3(config)#line vty 0 15
ACCESS_3(config-line)#transport input ssh
ACCESS_3(config-line)#login local
ACCESS_3(config-line)#exit
ACCESS_3(config)#exit
ACCESS_3#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
ACCESS_3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
ACCESS_3#
```

```
ACCESS_4#show ip shh
^
% Invalid input detected at '^' marker.

ACCESS_4#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
ACCESS_4#
```

```
ACCESS_5(config)#
*Mar 1 14:52:34.268: %SSH-5-ENABLED: SSH 1.99 has been enabled
ACCESS_5(config)#
ACCESS_5(config)#ip ssh version 2
ACCESS_5(config)#line vty 0 15
ACCESS_5(config-line)#transport input ssh
ACCESS_5(config-line)#login local
ACCESS_5(config-line)#end
ACCESS_5#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
ACCESS_5#
ACCESS_5#
ACCESS_5#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
ACCESS_5#
```

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configuración final de cada Switch

CORE

```
CORE_1>enable
CORE_1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	PRODUCCION	active	
20	DESARROLLO	active	
30	QA	active	
40	INFRA	active	
50	APP	active	
60	CIBER	active	
70	FINANZAS	active	
80	AUDITORIA	active	
90	MARKETING	active	
100	RRHH	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
CORE_1#
CORE_1#
CORE_1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	1

```
Port
Po1
```

Vlans allowed on trunk
1-1005

```
Port
Po1
```

Vlans allowed and active in management domain
1,10,20,30,40,50,60,70,80,90,100

```
Port
Po1
```

Vlans in spanning tree forwarding state and not pruned
1,10,20,30,40,50,60,70,80,90,100

```
CORE_1#
```

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```

CORE_1#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone   s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP       Gig0/1(P) Gig0/2(P)
CORE_1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
    Root ID    Priority    24577
              Address    00D0.BA03.2CA1
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
              Address    00D0.BA03.2CA1
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----+-----+-----+-----
Po1            Desg FWD 3         128.37 P2p

VLAN0010
  Spanning tree enabled protocol ieee
    Root ID    Priority    24586
              Address    00D0.BA03.2CA1
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    24586 (priority 24576 sys-id-ext 10)
              Address    00D0.BA03.2CA1
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
--More-- |

```

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DIS

```
DIS_2>enable
DIS_2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/4, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
10	PRODUCCION	active	
20	DESARROLLO	active	
30	QA	active	
40	INFRA	active	
50	APP	active	
60	CIBER	active	
70	FINANZAS	active	
80	AUDITORIA	active	
90	MARKETING	active	
100	RRHH	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
DIS_2#
DIS_2#
DIS_2#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	1
Fa0/1	on	802.1q	trunking	1
Fa0/3	auto	n-802.1q	trunking	1
Fa0/5	auto	n-802.1q	trunking	1

```
Port      Vlans allowed on trunk
Po1        1-1005
Fa0/1      1-1005
Fa0/3      1-1005
Fa0/5      1-1005

Port      Vlans allowed and active in management domain
Po1        1,10,20,30,40,50,60,70,80,90,100
Fa0/1      1,10,20,30,40,50,60,70,80,90,100
Fa0/3      1,10,20,30,40,50,60,70,80,90,100
Fa0/5      1,10,20,30,40,50,60,70,80,90,100

Port      Vlans in spanning tree forwarding state and not pruned
Po1        1,10,20,30,40,50,60,70,80,90,100
Fa0/1      1,10,20,30,40,50,60,70,80,90,100
Fa0/3      1,10,20,30,40,50,60,70,80,90,100
Fa0/5      1,10,20,30,40,50,60,70,80,90,100
```

```
DIS_2#
```

DIS_2#show etherchannel summary

Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
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-----+-----+-----+-----

1	Pol(SU)	LACP	Gig0/1(P) Gig0/2(P)
---	---------	------	---------------------

DIS_2#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	24577	
	Address	00D0.BA03.2CA1	
	Cost	3	
	Port	27 (Port-channel1)	
	Hello Time	2 sec	Max Age 20 sec Forward Delay 15 sec

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)
	Address	0001.634E.0DC8	
	Hello Time	2 sec	Max Age 20 sec Forward Delay 15 sec
	Aging Time	20	

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

-----+-----+-----+-----+-----+-----

Pol	Root	FWD	3	128.27	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/5	Desg	FWD	19	128.5	P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID	Priority	24586	
	Address	00D0.BA03.2CA1	
	Cost	3	
	Port	27 (Port-channel1)	
	Hello Time	2 sec	Max Age 20 sec Forward Delay 15 sec

Bridge ID	Priority	32778	(priority 32768 sys-id-ext 10)
-----------	----------	-------	--------------------------------

--More-- |

ACCESS_3

```
ACCESS_3>enable
ACCESS_3#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	PRODUCCION	active	Fa0/10
20	DESARROLLO	active	Fa0/11
30	QA	active	Fa0/12
40	INFRA	active	
50	APP	active	
60	CIBER	active	
70	FINANZAS	active	
80	AUDITORIA	active	
90	MARKETING	active	
100	RRHH	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ACCESS_3#
```

```
ACCESS_3#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005

Port	Vlans allowed and active in management domain
Fa0/1	1,10,20,30,40,50,60,70,80,90,100

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	1,10,20,30,40,50,60,70,80,90,100

```

ACCESS_3#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

```

```

Number of channel-groups in use: 0
Number of aggregators:          0

```

Group	Port-channel	Protocol	Ports
-------	--------------	----------	-------

```

ACCESS_3#show spanning-tree

```

```

VLAN0001

```

```

Spanning tree enabled protocol ieee

```

```

Root ID    Priority    24577
           Address    00D0.BA03.2CA1
           Cost        22
           Port        1(FastEthernet0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

```

```

Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0010.11D8.300D
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Fa0/1	Root	FWD	19	128.1	P2p
-------	------	-----	----	-------	-----

```

VLAN0010

```

```

Spanning tree enabled protocol ieee

```

```

Root ID    Priority    24586
           Address    00D0.BA03.2CA1
           Cost        22
           Port        1(FastEthernet0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

```

```

Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)

```

ACCESS_4

```
ACCESS_4>enable
ACCESS_4#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
40	VLAN0040	active	Fa0/10
50	VLAN0050	active	Fa0/11
60	VLAN0060	active	Fa0/12
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ACCESS_4#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/3	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/3	40,50,60

Port	Vlans allowed and active in management domain
Fa0/3	40,50,60

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/3	40,50,60

ACCESS_4#show etherchannel summary

Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 0

Number of aggregators: 0

Group	Port-channel	Protocol	Ports
-------	--------------	----------	-------

ACCESS_4#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	24577		
	Address	00D0.BA03.2CA1		
	Cost	22		
	Port	3(FastEthernet0/3)		
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)	
	Address	0004.9A3A.4501		
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec
	Aging Time	20		

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Fa0/3	Root	FWD	19	128.3	P2p
-------	------	-----	----	-------	-----

VLAN0040

Spanning tree enabled protocol ieee

Root ID	Priority	24616		
	Address	00D0.BA03.2CA1		
	Cost	22		
	Port	3(FastEthernet0/3)		
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec

Bridge ID	Priority	32808	(priority 32768 sys-id-ext 40)	
-----------	----------	-------	--------------------------------	--

--More-- |

ACCESS_5

```
ACCESS_5>enable
ACCESS_5#show vlan bri
ACCESS_5#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
70	VLAN0070	active	Fa0/10
80	VLAN0080	active	Fa0/11
90	VLAN0090	active	Fa0/12
100	VLAN0100	active	Fa0/13
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
ACCESS_5#show in
```

```
ACCESS_5#show interfaces tru
```

```
ACCESS_5#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/5	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/5	70,80,90,100

Port	Vlans allowed and active in management domain
Fa0/5	70,80,90,100

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/5	70,80,90,100

```
ACCESS_5#show et
```

```
ACCESS_5#show etherchannel sum
```

```
ACCESS_5#show etherchannel summary
```

```
Flags: D - down          P - in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```

ACCESS_5#show et
ACCESS_5#show etherchannel sum
ACCESS_5#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

```

```

Number of channel-groups in use: 0
Number of aggregators:          0

```

```

Group  Port-channel  Protocol  Ports
-----+-----+-----+-----

```

```

ACCESS_5#sho
ACCESS_5#show spann
ACCESS_5#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00D0.BA03.2CA1
             Cost        22
             Port        5(FastEthernet0/5)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0002.4A91.E050
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

```

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/5          Root FWD 19        128.5   P2p

```

```

VLAN0070
  Spanning tree enabled protocol ieee
  Root ID    Priority    24646
             Address     00D0.BA03.2CA1
             Cost        22
             Port        5(FastEthernet0/5)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

```

```

  Bridge ID  Priority    32838  (priority 32768 sys-id-ext 70)

```

```

--More-- |

```